

ANAMORPHIC ENCRYPTION: PRIVATE COMMUNICATION AGAINST A DICTATOR

BYPASSING COERCION IN MODERN CRYPTO SYSTEMS

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THE PROBLEM: CRYPTO UNDER COERCION

Traditional Cryptography's Vulnerable Assumptions:

- **Receiver-Privacy:** Assumes that a receiver's private key remains secure.
- **Sender-Freedom:** Assumes sender is free to choose messages.

The Dictator's Challenge:

- Coercive governments can force key revelation, which undermines receiver-privacy.
- Can force senders to transmit specific, chosen messages, undermining sender-freedom.
- This may outlaw alternative cryptosystems.

The Core Problem: How do we ensure private communication when fundamental cryptographic assumptions are legally and forcefully validated?

Motivation: To technically demonstrate the futility of government attempts to control strong cryptography.



WHAT IS ANAMORPHIC ENCRYPTION?

Anamorphic Encryption: a special encryption scheme whose public keys can be generated in one of two modes: normal or anamorphic.

Analogy: Like a distorted drawing that appears normal when viewed from a specific point or with a suitable mirror or lens.

Core Idea: Two messages in One Ciphertext:

- **Overt Message (m_0):** The normal message, visible to the dictator upon decryption with the normal secret key.
- **Covert/Anamorphic Message (m_1):** The hidden, special message that's only visible to parties with the double secret key.

Key Requirements for Plausibility & Security:

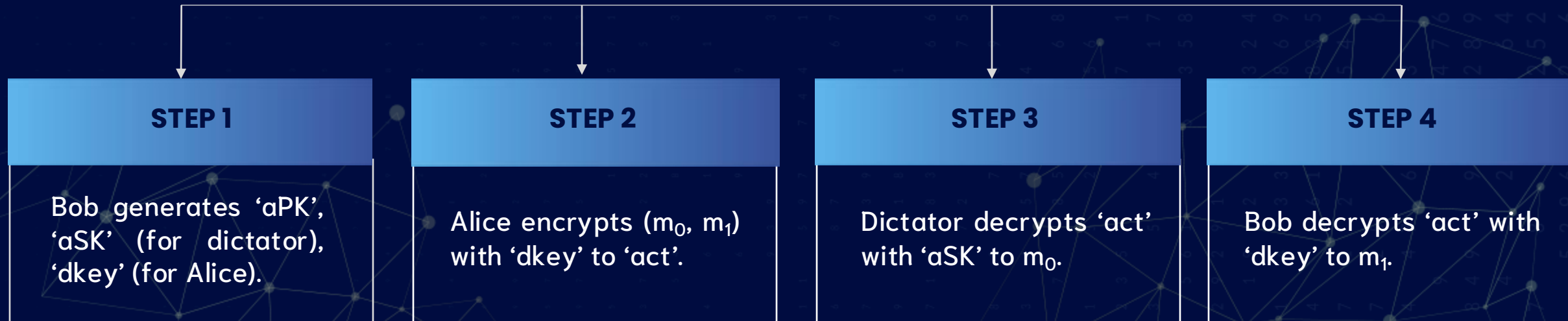
- Anamorphic public/secret key pairs must be indistinguishable from normal corresponding pairs.
- Ciphertexts produced using an anamorphic public key must be indistinguishable from those produced by a normal public key.

Goals: Zero-latency communication and high anamorphic bandwidth rate.

RECEIVER-ANAMORPHIC ENCRYPTION (COMBATting KEY SURRENDER)

GOAL: “To protect a receiver’s private message (m_1), even if the secret key is surrendered.”

Naor–Yung Basis Mechanism



Key Feature:

Simulated NIZK makes ‘act’ appear normal

Benefits:

Bandwidth Rate 1, Zero-Latency

SENDER-ANAMORPHIC ENCRYPTION

GOAL: “Alice sends (m_1) to Bob, while appearing to send (m_0) to Carol, even if coerced.”



Key Idea:

No prior shared secret between Alice and Bob needed.

Lattice-Based Mechanism:



Alice uses ‘fRandom’ (coin-toss faking algorithm)



Generates randomness R^* for $ct = \text{Enc}(\text{fPK}, m_0; R^*)$



ct decrypts to m_0 (with Carol’s fSK) and m_1 (with Bob’s dSK)

Requirements: “Common randomness,” “message recovery from randomness,” “equal distribution of plaintexts.”

SECURITY AND INDISTINGUISHABILITY

CORE PRINCIPLE: “Dictator cannot distinguish between ‘normal’ and ‘fully anamorphic’”

FORMAL PROOF (VIA GAMES)

‘NormalGame’ vs. ‘FullyAGame’

Probably of distinguishing is negligible.

IMPLICATIONS

Secrecy of m_i : Covert message remains hidden

Plausibility: Operations appear legitimate; no suspicion raised.

UNDERLYING TOOLS

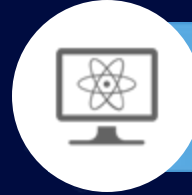
Underlying Tools: IND-CPA secure PKE and Simulation Sound NIZK.

PRACTICALITY AND LIMITATIONS

ADVANTAGES



- Leverages Existing Cryptosystems.
- Zero-Latency Communication
- High Bandwidth Rate (Receiver-AM)
- No Shared Secret (Sender-AM)



CHALLENGES

- Simple Rejection Sampling is Inefficient
- Shared 'dkey' for Receiver-AM (can be a physical exchange).
- Not all PKEs are naturally Anamorphic (Ex. Goldwasser-Micali for Sender-AM).
- Potential "Crypto Wars" of banning specific schemes

CONCLUSION

SUMMARY

- ▶ Anamorphic Encryption demonstrates the futility of government control over strong crypto.

IMPACT

- ▶ Empowers private communication even under extreme duress.

FUTURE

- ▶ Explore anamorphic properties in more cryptosystems
- ▶ Combine Anamorphism with other covert techniques (steganography, kleptography)
- ▶ Analyze broader policy and societal implications.

THANK YOU